Lecture 11: Testing Multiple Equations and Review of Inference

Jeremy Majerovitz (University of Notre Dame)

Econometrics, Fall 2026

Exercises

Suppose we are studying the effect of a training program on income. Participants are randomly assigned to one of three groups: control (no training), basic training, or advanced training. We estimate the following regression:

$$\text{Monthly Income}_i = \alpha + \beta_1 \text{Basic}_i + \beta_2 \text{Advanced}_i + \varepsilon_i$$

where Basic_i is a dummy variable equal to one if the individual was assigned to basic training, and Advanced_i is a dummy variable equal to one if the individual was assigned to advanced training.

Exercise 1. Which group is the omitted category? What difference in means does β_1 represent? What difference in means does β_2 represent?

Exercise 2. Suppose I want to test whether basic training affects income. What is my null hypothesis? How would I construct a 5% test of this null? Set up the t-statistic and state the appropriate critical value.

Exercises

Suppose we are studying the effect of a training program on income. Participants are randomly assigned to one of three groups: control (no training), basic training, or advanced training. We estimate the following regression:

$$\text{Monthly Income}_i = \alpha + \beta_1 \text{Basic}_i + \beta_2 \text{Advanced}_i + \varepsilon_i$$

where Basic_i is a dummy variable equal to one if the individual was assigned to basic training, and Advanced_i is a dummy variable equal to one if the individual was assigned to advanced training.

Exercise 3. Suppose I want to test whether basic training and advanced training have the same effect. What is my null hypothesis? How would I construct a 5% test of this null, using the linear combination approach that we outlined in class? How would I implement that approach in Stata?

Exercises

Suppose we are studying the effect of a training program on income. Participants are randomly assigned to one of three groups: control (no training), basic training, or advanced training. We estimate the following regression:

$$\text{Monthly Income}_i = \alpha + \beta_1 \text{Basic}_i + \beta_2 \text{Advanced}_i + \varepsilon_i$$

where Basic_i is a dummy variable equal to one if the individual was assigned to basic training, and Advanced_i is a dummy variable equal to one if the individual was assigned to advanced training.

Exercise 4. Suppose I want to test whether basic training and advanced training have the same effect, but I do not want to use the linear combination approach. How could I change the dummy variables in my regression so that one of the coefficients reflects the difference between the basic and advanced training?

Exercises

Suppose we are studying the effect of a training program on income. Participants are randomly assigned to one of three groups: control (no training), basic training, or advanced training. We estimate the following regression:

$$\text{Monthly Income}_i = \alpha + \beta_1 \text{Basic}_i + \beta_2 \text{Advanced}_i + \varepsilon_i$$

where Basic_i is a dummy variable equal to one if the individual was assigned to basic training, and Advanced_i is a dummy variable equal to one if the individual was assigned to advanced training.

Exercise 5. Returning to the original form of the regression, suppose that I want to test the hypothesis that neither training has any effect. State that null hypothesis. How would I test that null? How would I implement that test in Stata?

Exercises

We estimate the following regression:

$$\text{Monthly Income}_i = \alpha + \beta_1 \text{Basic}_i + \beta_2 \text{Advanced}_i + \varepsilon_i$$

where Basic_i is a dummy variable equal to one if the individual was assigned to basic training, and Advanced_i is a dummy variable equal to one if the individual was assigned to advanced training.

Exercise 6. Suppose that I get the estimates $\hat{\beta}_1 = 60$, $\widehat{\text{SE}}\left(\hat{\beta}_1\right) = 50$, $\hat{\beta}_2 = 90$, and $\widehat{\text{SE}}\left(\hat{\beta}_2\right) = 40$. Can we reject, at the 5% level, that basic training has no effect on monthly income? What about advanced training? Can we conclude that basic training has no effect?

Exercise 7. Under the above estimates, what can I say about the effects of basic versus advanced training? Can I use the information above to test whether advanced training is more effective than basic training, at the 5% level?